

Pr-2 Data Cleaners

classmate

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Q.1.6 Data preprocessing

→ Data pre processing is the process of cleaning and transforming raw data into a suitable format for analysis and machine learning.

Q.2.1 Missing Values

→ Missing Values are values that are not available or recorded in a dataset.

Q.3.1 Mean Imputation

→ Mean imputation replaces missing numerical value with the average of the available values.

Q.4.1 Median Imputation

→ Median Imputation replaces missing numerical ~~value~~ value with the middle value of the data and is less affected by outliers than the mean.

Q.5.1 Most frequent Imputation

→ Most frequent imputation replaces missing value with the value that occurs most often. It is commonly used for categorical data.

Q.6.1 Random Sample Imputation

→ Random Sample Imputation replaces missing values with randomly selected values from existing non-missing observations.

Q.7.1 KNN Imputation

→ KNN imputation estimates missing values using similar observations, called nearest neighbours.

Q.8) MICE

→ MICE means Multiple Imputation by chained Equations. It estimates missing value using relationships between different variables.

Q.9) outliers

→ An outlier is a value that is significantly different from most other observations.

Q.10) Z-Score.

→ The Z-score method measures how far a value is from the mean in terms of standard deviations.

Q.11) IQR

→ IQR stands for Interquartile Range and represents the spread of the middle 50% of the data.
formula → $IQR = Q_3 - Q_1$.

Q.12) Percentile Method

→ The Percentile method identifies extreme observation using lower and upper percentile limits.

Q.13) winsorization.

→ winsorization reduces the effect of extreme value by replacing them with selected boundary values instead of deleting records.

Q.14) Importance of data cleaning.

→ Data cleaning improves data quality and prepares the dataset for reliable analysis and machine learning.

Q1st Target variable outlier features

→ The target variable 'bmi', 'blood pressure', 'cholesterol', and 'glucose' for outliers.

Q2nd Why Compare Multiple Methods?

→ Different techniques can produce different results. Comparing them helps select a suitable method for the final dataset.

Q3rd final Goal

→ The goal is to handle missing values and outliers and create dataset suitable for machine learning.